

Joris Koefer

RESEARCH INTEREST

Positive Geometry and its applications in QFTs

EDUCATION

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| 2023 - present | MPI for Mathematics in the Sciences, Leipzig, GER <ul style="list-style-type: none">• PhD in Mathematics<ul style="list-style-type: none">◦ Non-linear algebra group◦ Advisor: Rainer Sinn |
| 2022 - 2023 | University of Warwick, Coventry, UK <ul style="list-style-type: none">• M.Sc. in Mathematics (1-year) - First Class<ul style="list-style-type: none">◦ Advisor: Diane Maclagan◦ Thesis: A computational approach for tropical solutions to enumerative problems |
| 2020 - 2022 | Ruhr University Bochum, Bochum, GER <ul style="list-style-type: none">• B.Sc in Mathematics (3-year) - Excellency Award<ul style="list-style-type: none">◦ Advisor: Christian Stump◦ Thesis: On combinatorial models for the positive Grassmannian with a view towards Amplituhedra |
| 2018 - 2020 | Ruhr University Bochum, Bochum, GER <ul style="list-style-type: none">• B.Sc. in Physics (3-year) - First Class<ul style="list-style-type: none">◦ Advisor: Evgeny Epelbaum◦ Thesis: Application of a subtractive renormalisation scheme in Nucleon-Nucleon scattering in Effective Field Theory |
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TEACHING

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| 2019 - 2022 | Ruhr University Bochum, Bochum, GER <ul style="list-style-type: none">• Teaching Assistant<ul style="list-style-type: none">◦ Mathematical Methods for Physicists I + II◦ Classical Theoretical Physics |
| 2019 - 2021 | Ruhr University Bochum, Bochum, GER <ul style="list-style-type: none">• Lab Supervisor<ul style="list-style-type: none">◦ Biophysics◦ Geophysics |

Research

- 2025** **Connection Matrices in Macaulay2**
With Paul Görlach, Anna-Laura Sattelberger, Mahrud Sayrafi, Hendrik Schroeder, Nicolas Weiss, Francesca Zaffalon
[arXiv](#)
- 2025** **Taking the Amplituhedron to the limit**
With R. Sinn
[arXiv](#)
- 2024** **Uniqueness of MHV Gravity Amplitudes**
With U.Oktem, S. Paranjape, J. Trnka and B. Zacovic
To appear in Le Matematiche - Special Volume on Positive Geometry
[arXiv](#)

TALKS & PRESENTATIONS

- 2025** **Taking the Amplituhedron to the limit**, UiO Oslo
- 2024** **Positive Geometries**, MPI MIS Leipzig
- Meeting of Scientific Advisory Board, [Slides](#)
- 2022** **Combinatorial Models of the Grassmannian**
- Seminar on Combinatorics, Ruhr University Bochum, [Link](#)